Mammalian Genome 7, 52–54 (1996).

Mammalian Genome

© Springer-Verlag New York Inc. 1996

A major locus (*RN*) affecting muscle glycogen content is located on pig Chromosome 15

P. Mariani,[1] K. Lundström,[2] U. Gustafsson,[1] A.-C. Enfält,[2] R.K. Juneja,[1] L. Andersson[1]

[1]Department of Animal Breeding and Genetics, Swedish University of Agricultural Sciences, BMC, Box 597, S-751 24 Uppsala, Sweden
[2]Department of Food Science, Swedish University of Agricultural Sciences, Uppsala, Sweden

Received: 18 April 1995 / Accepted: 16 June 1995

Abstract. The *RN* locus in pigs has a major effect on the amount of stored glycogen in white muscle and affects meat quality. The fully dominant RN^- allele, associated with high glycogen content, occurs in the Hampshire breed. We have mapped the *RN* locus using a large half-sib family comprising one heterozygous RN^-/rn^+ Hampshire boar mated to homozygous rn^+/rn^+ Swedish Landrace × Swedish Yorkshire sows. The segregation at the *RN* locus was inferred from data on glycolytic potential and residual glycogen in white muscle which both showed clear bimodal distributions. Highly significant evidence for genetic linkage was obtained against microsatellite markers on Chromosome (Chr) 15. Multipoint analysis revealed the order *Sw1111*–8.0–*S0088*–10.6–*RN*–4.8–*Sw936,Sw906* (recombination estimates are given as Kosambi cM). Comparative mapping data imply that the human homolog of *RN* is located on Chr 2q.

Introduction

A major locus with two alleles, RN^- and rn^+, having a major effect on meat quality in pigs has been identified in the Hampshire breed by family segregation analysis (Le Roy et al. 1990). In comparison with other breeds, the musculature of purebred Hampshire pigs has a higher glycogen content and lower ultimate pH (Monin and Sellier 1985; Naveau et al. 1985). This difference could to a large extent be explained by the presence of the fully dominant RN^- allele, which increases the amount of glycogen in muscle by as much as 70% (Estrade et al. 1993a, 1993b). The high glycogen content is probably due to an anomalous glycogen metabolism in RN^- carriers (Estrade et al. 1994). The large phenotypic difference between $RN^-/-$ and rn^+/rn^+ animals makes it possible to score the segregation at the *RN* locus as a monogenic trait. We have utilized this possibility in an attempt to map the *RN* locus by use of microsatellite markers (Ellegren et al. 1994; Rohrer et al. 1994; Archibald et al. 1995). We report here that the *RN* locus is located on the distal part of Chr 15.

Materials and methods

Animals. One large half-sib family comprising one Hampshire boar, 22 Swedish Landrace × Swedish Yorkshire sows, and 105 male offspring were used. The sire was heterozygous RN^-/rn^+ while all dams were homozygous rn^+/rn^+. The pigs were slaughtered at an average live weight of 106 kg, and blood samples were collected. DNA was prepared from all animals passing a conventional parentage test based on nine blood protein polymorphisms (Gahne and Juneja 1985; Juneja and Gahne 1987).

Classification of phenotypic traits. Muscle samples were taken from *Musculus longissimus dorsi* (LD) 4 days after slaughter and frozen for subsequent analyses. The glycolytic potential (GP) was determined according to Monin and Sellier (1985) and Talmant and associates (1989), as GP = 2 × ([glycogen] + [glucose] + [glucose-6-phosphate]) + [lactate], and expressed as μmol lactate equivalents per gram muscle wet weight. The *RN* genotypes were also classified by use of the sum of glycogen, glucose, and glucose-6-phosphate in μmol/g muscle wet weight; this sum will be referred to as residual glycogen (GL_{Res}).

Genetic markers. Markers informative for linkage analyses against the *RN* locus were identified by screening a large number of previously described porcine microsatellites (Ellegren et al. 1994; Rohrer et al. 1994; Archibald et al. 1995). All markers for which the single Hampshire boar was heterozygous were tested for cosegregation against the *RN* locus. The initial set comprised the following markers assigned to respective chromosome; Chr 1, *S0020;* Chr 3, *Sw72, S0164, Sw1066;* Chr 4, *S0097, Sw835;* Chr 5, *Sw995;* Chr 7, *S0013, S0029, Sw632;* Chr 8, *SPP1;* Chr 9, *S0019, APOA1;* Chr 13, *Sw38, Sw935;* Chr 14, *Sw342;* Chr 15, *Sw936;* Chr 17, *Sw840;* Chr 18, *Sw787.* The sire was also heterozygous at the *A1BG* locus (Chr 6), encoding one of the protein polymorphisms utilized in the parentage test. Subsequent to the finding of linkage to *Sw936* on Chr 15, three additional microsatellites (*S0088, Sw906,* and *Sw1111*) from this chromosome were scored in the pedigree. Information on all markers have been reported previously (Ellegren et al. 1994; Rohrer et al. 1994; Archibald et al. 1995).

PCR amplification of microsatellite markers was carried out with a thermocycler (Techne PHC3) in a total volume of 10 μl with 100 ng genomic DNA, 1.5 pmol end-labeled primer, and 0.5 units Taq-polymerase (Dynazyme, Finnzymes). A prolonged denaturation step (92°C for 2 min) was followed by 30 cycles of 94°C for 30 s, 58°C for 30 s, and 72°C for 30 s. The PCR products were separated by electrophoresis in 6% denaturing polyacrylamide gels and visualized by autoradiography.

Linkage analysis. Two-point and multi-point lod score analyses were performed with the CRI-MAP, version 2.4, linkage package (Green et al. 1990) on a Sun SPARC-2 workstation (SunOS rel. 4.2.1). A lod score value above 3 was considered as significant evidence for linkage. The multi-point map was constructed with the BUILD option, and the statistical support for the most likely order was assessed with the FLIPS option. Recombination events were identified by use of CHROMPIC. Observed recombination fractions were transformed to map distances by use of the Kosambi map function.

Results and discussion

The two phenotypic traits, glycolytic potential (GP) and the sum of glycogen, glucose, and glucose-6-phosphate (GL_{Res}), both showed clear bimodal distributions (Fig. 1a and b). Animals with a GP value above 180 μmol/g were assumed to be heterozygous RN^-/rn^+, while animals below this threshold were considered to be rn^+/rn^+. Similarly, animals with a GL_{Res} value above 40 μmol/g were classified as RN^-/rn^+. There was an excellent correlation between the two traits, and there was a discrepancy in classification for only one animal. This animal had a GP value of 167

Correspondence to: L. Andersson

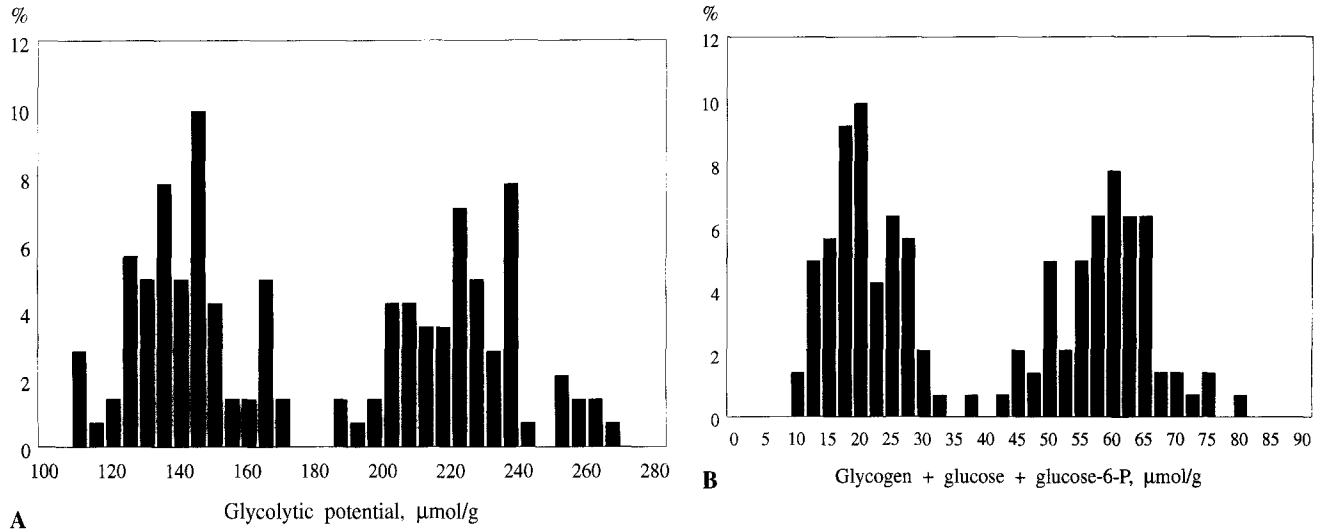


Fig. 1. Frequency distribution of **(A)** glycolytic potential and **(B)** the sum of glycogen, glucose, and glucose-6-phosphate in a group of half-sibs sired by a single RN^-/rn^+ heterozygous Hampshire boar mated to rn^+/rn^+ homozygous Swedish Yorkshire × Swedish Landrace sows. Individuals with a glycolytic potential >180 μmol/g and a sum of glycogen, glucose, and glucose-6-phosphate >40 μmol/g were classified as carriers of the RN^- gene.

μmol/g and a GL_{Res} value of 43 μmol/g. These values are in the upper and lower tail, respectively, of the two distributions. The *RN* genotype of this animal was consequently considered unknown.

The segregation at the *RN* locus was tested for genetic linkage against a set of 20 genetic markers representing 12 different autosomes. Highly significant two-point lod scores were obtained against the microsatellite *Sw936* on Chr 15. Three additional microsatellites (*S0088, Sw906,* and *Sw1111*) were subsequently included in the linkage analyses. The results of the two-point analyses are compiled in Table 1. Multipoint analyses showed with great confidence (lod score support above 3) that *RN* is located between *S0088* and *Sw936/Sw906* (Fig. 2).

There was excellent agreement between the *RN* genotype classification made from phenotypic data and the one deduced with linked markers. In fact, we observed no double recombinants, which would be expected to be found if we had misclassifications in the RN data. This means that all recombination events involving *RN* could be confirmed by recombination events between the flanking microsatellites. Thus, our RN classification with GP and GL_{Res} appears to be highly reliable, as suggested by the clear bimodal distributions of the two traits (Fig. 1). There was one animal where the RN classification by use of the two traits was inconsistent. The data on the linked markers strongly suggested that this animal was RN^-/rn^+ in agreement with the GL_{Res} data.

The four microsatellite markers used in this study to map the *RN* locus were also included in the Chr 15 map presented by Rohrer and colleagues (1994). The linear order of loci is in perfect agreement between the two studies. However, there is a clear difference as regards the map distance in the interval between *Sw1111* and *Sw906,* where we observed a figure of about 23 cM in contrast to an estimate above 50 cM in the previous study (Rohrer et al. 1994). This striking difference could be explained at least partly by sex differences in recombination rates, since we report a male-specific map while Rohrer and coworkers reported a sex-averaged map.

The *S0088* marker is closely linked (3 cM) to the locus for dipeptidylpeptidase IV (*DPP4*), which previously has been mapped by in situ hybridization to 15q2.1 (Thomsen et al. 1993). We can therefore conclude that RN is located in the region 15q2.1–15qtel. A more narrow physical localization is currently prohibited by the lack of physically mapped, polymorphic markers in this area.

The result of the present study will be the starting point for further mapping of the *RN* locus with the ultimate goal to clone the gene and characterize the mutation. One possible strategy for this will be to utilize information on the corresponding regions in the human and murine genome to identify possible candidate genes involved in glycogen metabolism or candidate loci with similar phenotypic effects as the *RN* locus. It is clear that this part of porcine Chr 15 shares homology with a part of human Chr 2q (Thomsen et al. 1993; Johansson et al. 1995; Rettenberger et al. 1995). This region harbors the *GCG* locus encoding the hormone

Table 1. Summary of pair-wise lod score analyses for the *RN* locus against markers on pig Chr 15.

Pair of loci	Recombination fraction	Lod score
RN–Sw1111	0.19	7.9
RN–S0088	0.10	15.7
RN–Sw936	0.05	21.0
RN–Sw906	0.06	15.9

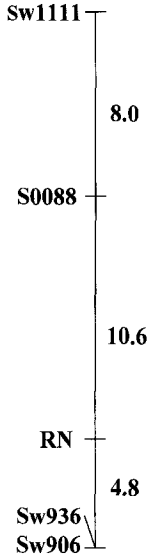


Fig. 2. Summary of the linkage map around the *RN* locus on pig Chr 15. The map distances are in Kosambi cM and represent male recombination rates from one single *RN* heterozygous boar.

glucagone and the *UGP2* locus coding for UDP-glucose pyrophosphorylase (O'Brien 1993), which both influence glycogen metabolism and therefore are putative candidate genes for the *RN* mutation.

The linkage between *RN* and *S0088* on pig Chr 15 has independently been found by Milan and coworkers (1995).

Acknowledgments. The authors are grateful to Sven and Kent Nilsson for providing data and animal material from their herd. We thank Ms. L. Lundkvist and Ms. M. Lundström for help with the biochemical analyses, and Dr. W. Coppetiers for his kind gift of microsatellite primers. We also gratefully acknowledge the contribution of microsatellite primers provided by Max Rothschild, US Pig Genome Coordinator, as part of the cooperative pig gene mapping efforts supported by the U.S. National Animal Genome Research Program NRSP-8. The work was financially supported by the Swedish Research Council for Forestry and Agriculture and by 'Stiftelsen Svensk Svinforskning'.

References

Archibald, A.L., et al. (1995). The PigMaP Consortium linkage map of the pig (*Sus scrofa*). Mamm. Genome 6, 157–175.

Ellegren, H., Chowdhary, B.P., Johansson, M., Marklund, L., Fredholm, M., Gustavsson, I., Andersson, L. (1994). A primary linkage map of the porcine genome reveals a low rate of genetic recombination. Genetics 137, 1089–1100.

Estrade, M., Vignon, X., Monin, G. (1993a). Effect of the RN^- gene on ultrastructure and protein fractions in pig muscle. Meat Sci. 35, 313–319.

Estrade, M., Vignon, X., Rock, E., Monin, G. (1993b). Glycogen hyperaccumulation in the white muscle fibres of RN^- carrier pigs. A biochemical and ultrastructural study. Comp. Biochem. Physiol. 104B(2), 321–326.

Estrade, M., Ayoub, S., Talmant, A., Monin, G. (1994). Enzyme activities of glycogen metabolism and mitochondrial characteristics in muscle of RN^- carrier pigs (*Sus scrofa domesticus*). Comp. Biochem. Physiol. 108B(3), 295–301.

Gahne, B., Juneja, R.K. (1985). Prediction of the Halothane (Hal) genotypes of pigs by deducing Hal, Phi, Po2, Pgd haplotypes of parents and offspring: results from a large-scale practice in Swedish breeds. Anim. Blood Groups Biochem. Genet. 16, 265–283.

Green, P., Falls, K., Crooks, S. (1990). Documentation for CRI-MAP, version 2.4. St. Louis: Washington University School of Medicine.

Johansson, M., Ellegren, H., Andersson, L. (1995). Comparative mapping reveals extensive linkage conservation—but with gene order rearrangements—between the pig and the human genomes. Genomics 25, 682–690.

Juneja, R.K., Gahne, B. (1987). Simultaneous phenotyping of pig plasma α-protease inhibitors (PI1, PO1A, PO1B, PI2) and four other proteins (PO2, TF, CP, HPX) by a simple method of 2D horizontal electrophoresis. Anim. Genet. 18, 197–211.

Le Roy, P., Naveau, J., Elsen, J.M., Sellier, P. (1990). Evidence for a new major gene influencing meat quality in pigs. Genet. Res. 55, 33–40.

Milan, D., Le Roy, P., Woloszyn, N., Caritez, J.C., Elsen, J.M., Sellier, P., Gellin, J. (1995). The RN locus for meat quality maps to pig chromosome 15. Genet. Sel. Evol. (Paris), in press.

Monin, G., Sellier, P. (1985). Pork of low technological quality with a normal rate of muscle pH fall in the immediate post-mortem period: the case of the Hampshire breed. Meat Sci. 13, 49–63.

Naveau, J., Pommeret, P., Lechaux, P. (1985). Proposition d'une méthode de mesure du rendement technologique: 'la méthode NAPOLE'. Techniporc 8, 7–13.

O'Brien, S.J., ed. (1993). *Genetic Maps,* 6th ed. (Cold Spring Harbor, N.Y.: Cold Spring Harbor Laboratory Press).

Rettenberger, G., Klett, C., Zechner, U., Kunz, J., Vogel, W., Hameister, H. (1995). Visualization of the conservation of synteny between pigs and humans by heterologous chromosomal painting. Genomics 26, 372–378.

Rohrer, G.A., Leeson, J.A., Keele, J.W., Smith, T.P., Beattie, C.W. (1994). A microsatellite linkage map of the porcine genome. Genetics 136, 231–245.

Talmant, A., Fernandez, X., Sellier, P., Monin, G. (1989). Glycolytic potential in Longissimus dorsi muscle of Large White pigs, as measured after in vivo sampling. Proc. 35th International Congress of Meat Science and Technology, Copenhagen, pp. 1129–1132.

Thomsen, P.D., Qvist, H., Marklund, L., Andersson, L., Sjöström, H., Norén, O. (1993). Assignment of the dipeptidylpeptidase IV (DPP4) gene to pig Chromosome 15q21. Mamm. Genome 4, 604–607.